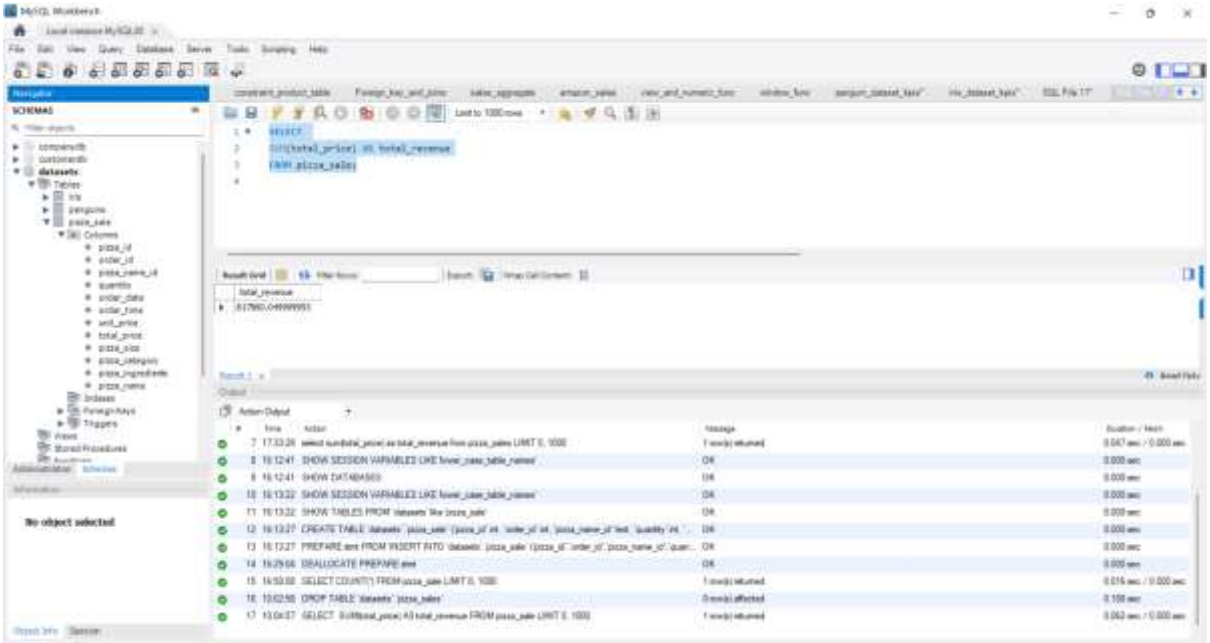
Pizza Sales Analysis Using SQL

KPI'S REQUIREMENT:

We need to analyze key indicators for our pizza sales data to gain insights into our business performance. Specifically, we want to calculate the following metrics.

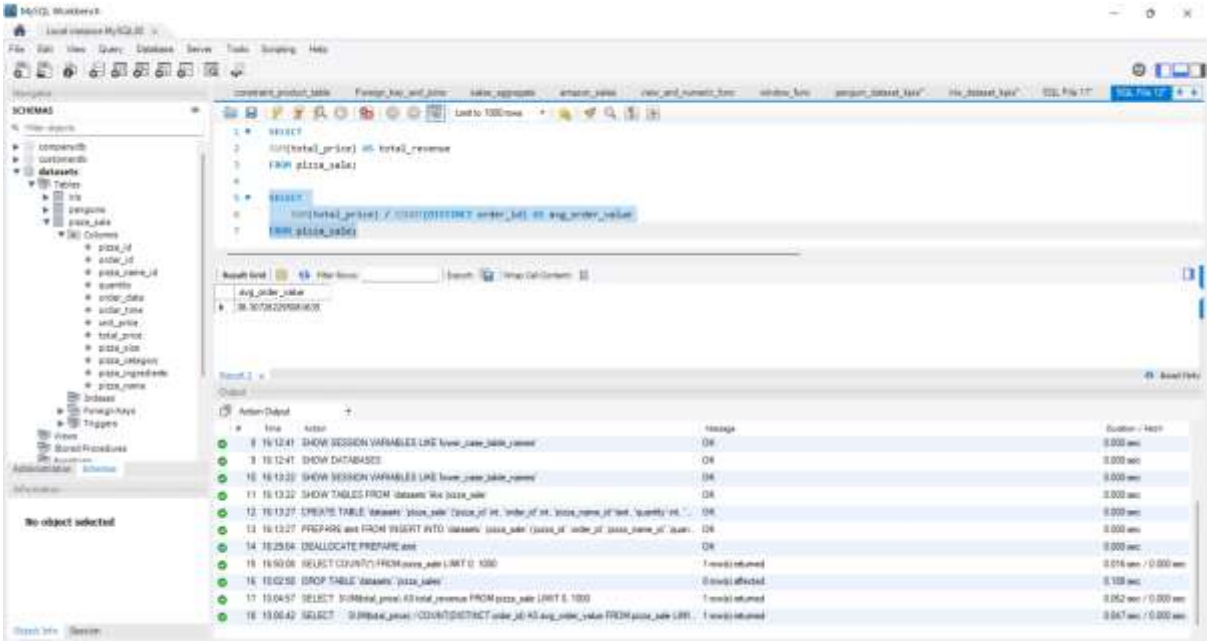
1. TOTAL REVENUE: The sum of the total price of all pizza orders.

```
SELECT
SUM(total_price) AS total_revenue
FROM pizza_sale;
```



2. AVERAGE ORDER VALUE: The average amount spent per order, calculated by dividing the total revenue by the total number of orders.

```
SELECT
   SUM(total_price) / COUNT(DISTINCT order_id) AS avg_order_value
FROM pizza_sale;
```

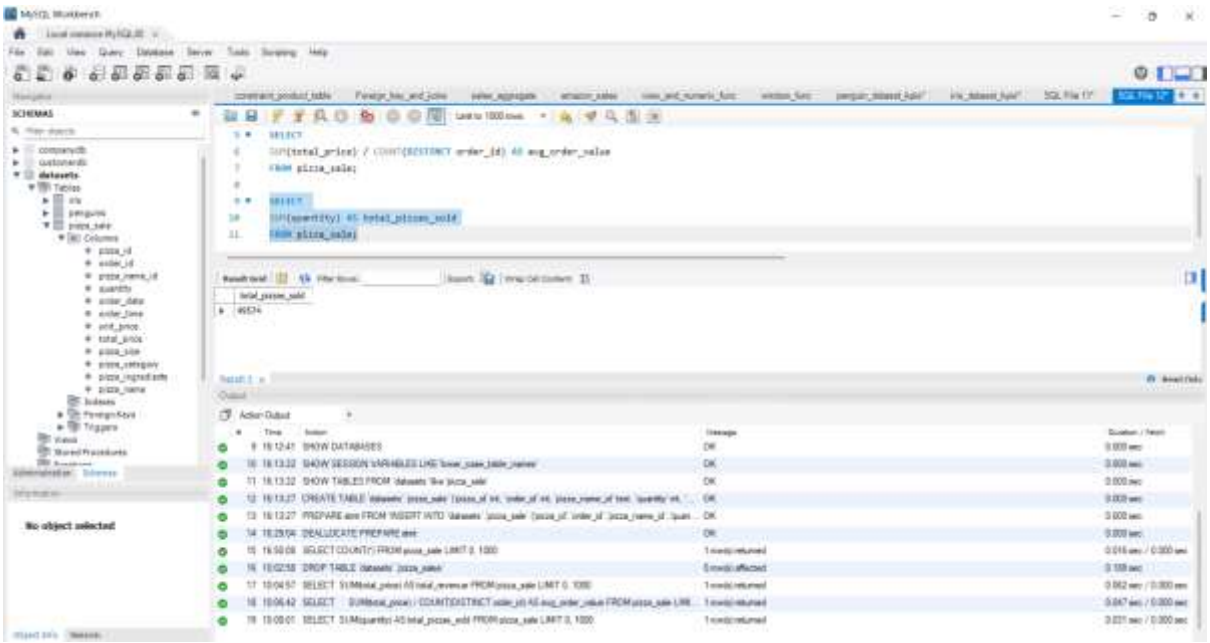


3. TOTAL PIZZA'S SOLD: The sum of the quantities of all pizzas sold

SELECT

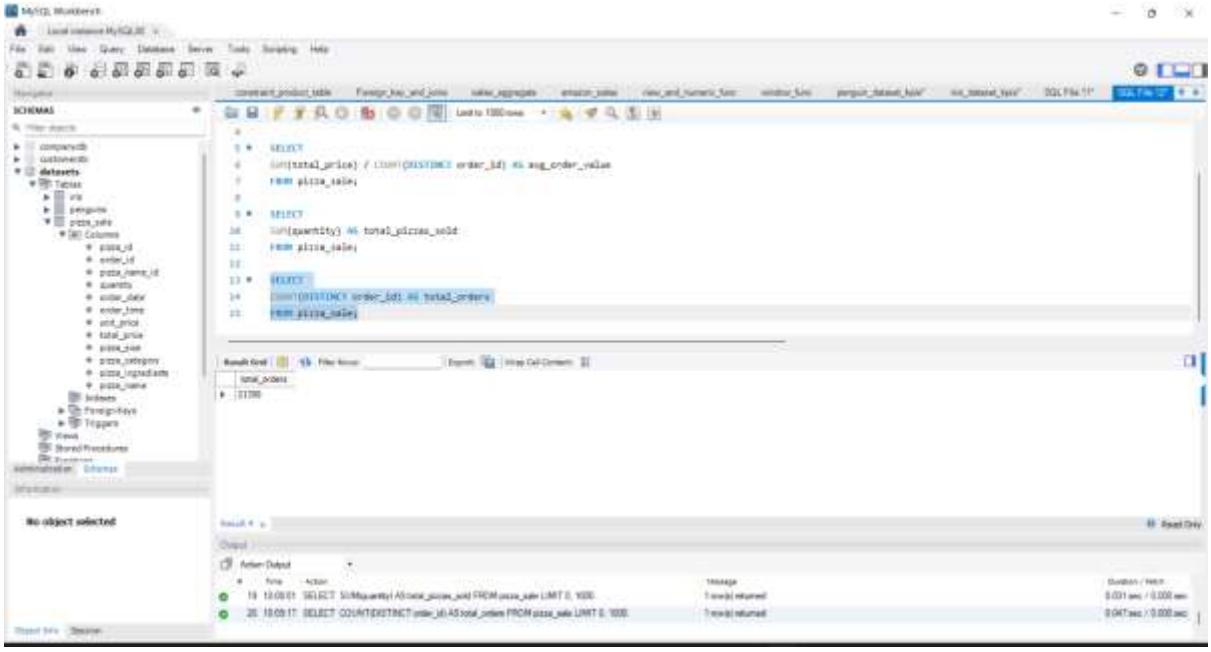
SUM(quantity) AS total_pizzas_sold

FROM pizza_sale;



4. TOTAL ORDERS: The total number of orders placed

```
SELECT
COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sale;
```

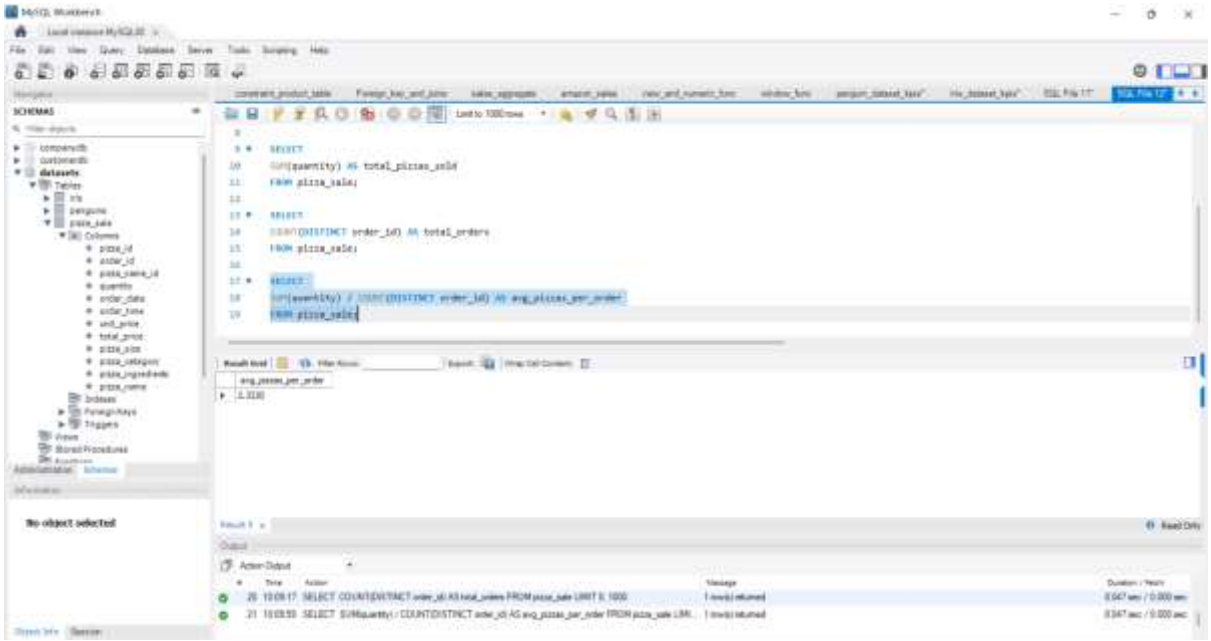


5. AVERAGE PIZZAS PER ORDER: The average number of pizzas sold per order, calculated by dividing the total number of pizzas sold by total number of orders.

```
SELECT

SUM(quantity) / COUNT(DISTINCT order_id) AS avg_pizzas_per_order

FROM pizza_sale;
```



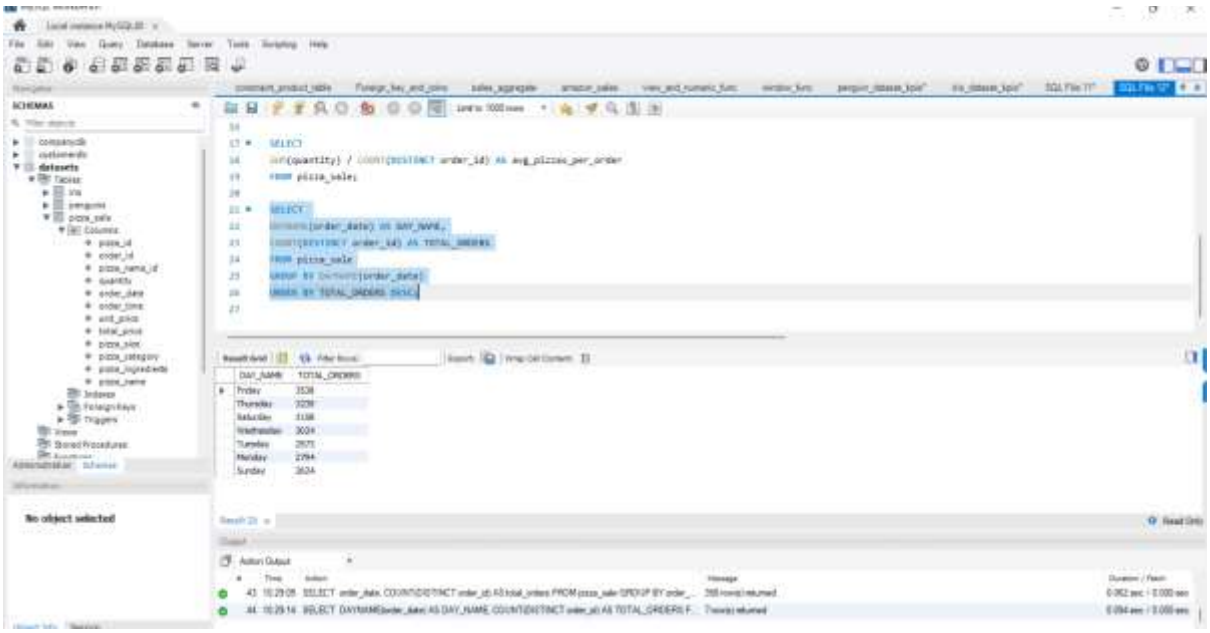
CHARTS REQUIREMENT:

We would like to visualize various aspects of our pizza sales data to gain insights and understand key trends. We have identified the following requirements for creating charts:

1.DAILY TREND FOR TOTAL ORDERS:

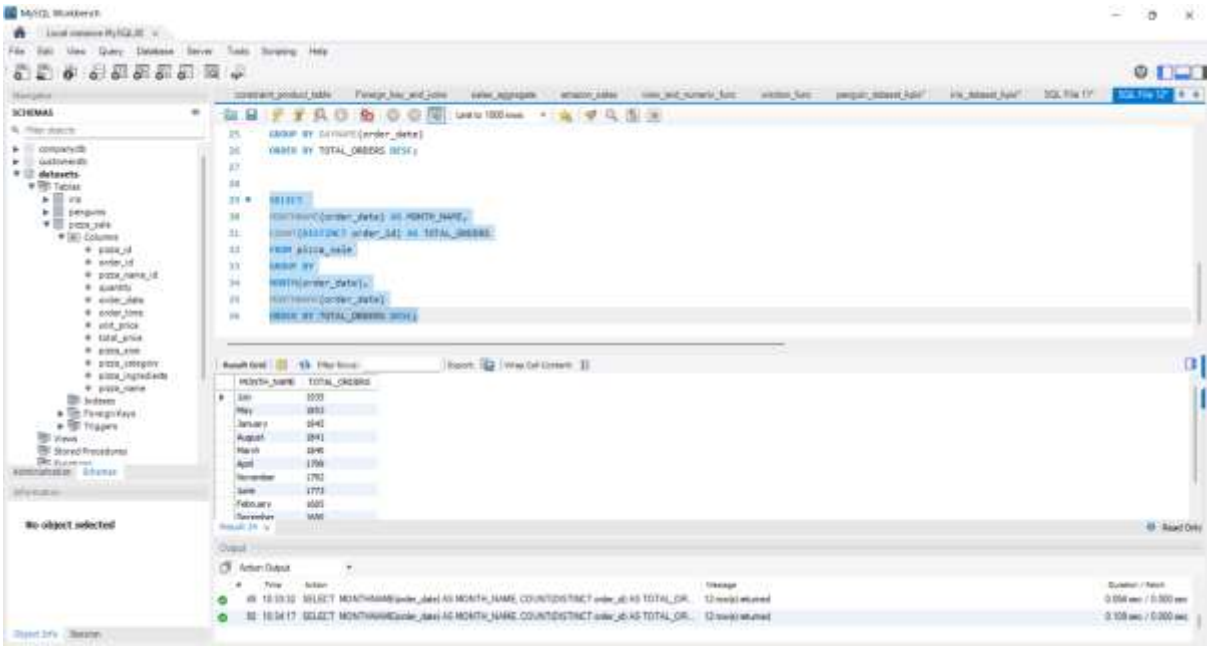
Create a bar chart that displays the daily trend of total orders over a specific time period. This chart will help us identity any patterns or fluctuations in order volumes on a daily basis.

```
SELECT
DAYNAME(order_date) AS DAY_NAME,
COUNT(DISTINCT order_id) AS TOTAL_ORDERS
FROM pizza_sale
GROUP BY DAYNAME(order_date)
ORDER BY TOTAL_ORDERS DESC;
```



2.MONTHLY TREND FOR TOTAL ORDERS:
Create a line chart that illustrates the hourly trend of total orders throughout the Month. This chart will allow you to identify peak months of high order activity.

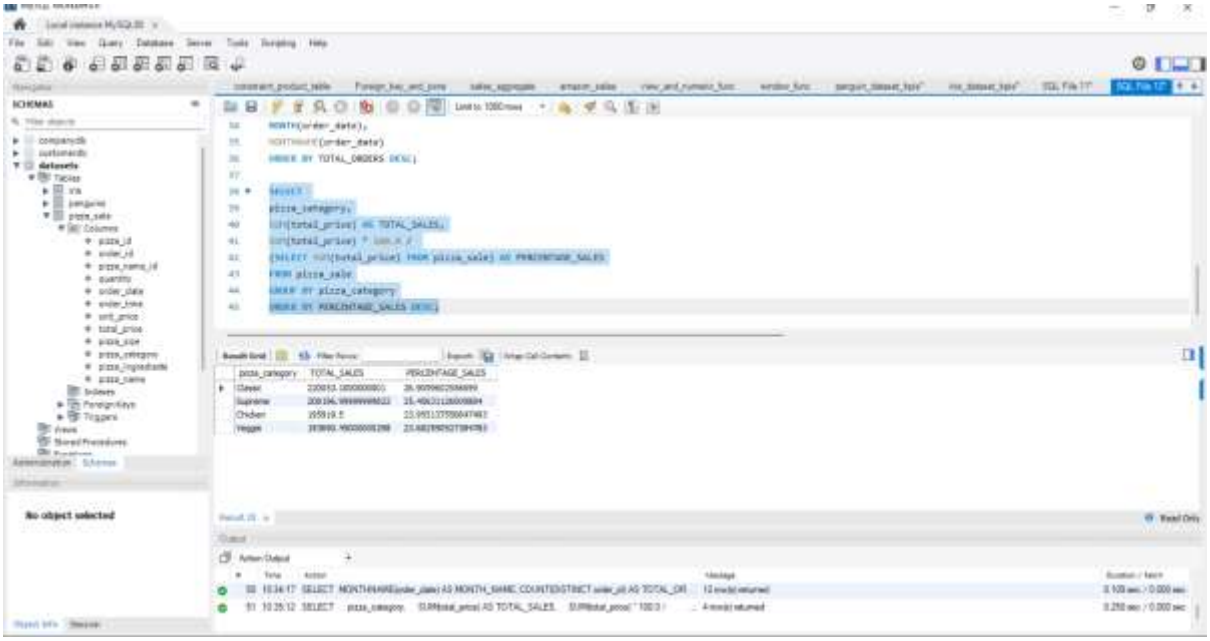
```
SELECT
MONTHNAME(order_date) AS MONTH_NAME,
COUNT(DISTINCT order_id) AS TOTAL_ORDERS
FROM pizza_sale
GROUP BY
MONTH(order_date),
MONTHNAME(order_date)
ORDER BY TOTAL_ORDERS DESC;
```



3.PERCENTAGE OF SALES BY PIZZA CATEGORY:

Create a pie chart that shows the distribution of sales across different pizza categories. This chart will provide insights into the popularity of various pizza categories and their contribution to overall sales.

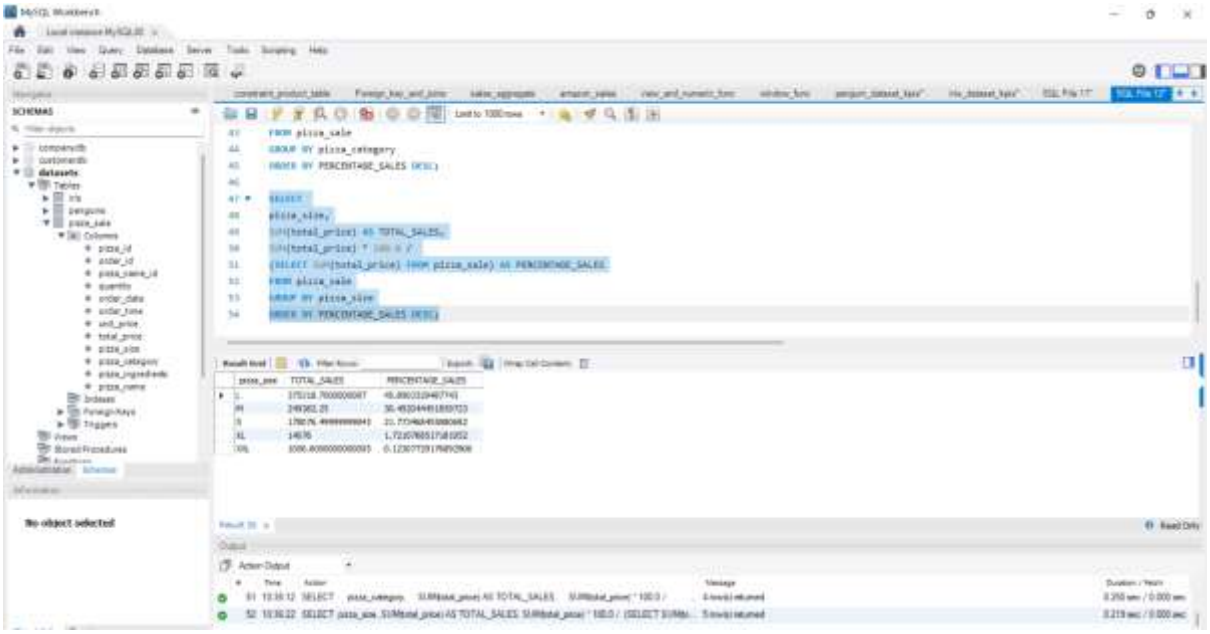
```
SELECT
pizza_category,
SUM(total_price) AS TOTAL_SALES,
SUM(total_price) * 100.0 /
(SELECT SUM(total_price) FROM pizza_sale) AS PERCENTAGE_SALES
FROM pizza_sale
GROUP BY pizza_category
ORDER BY PERCENTAGE_SALES DESC;
```



4.PERCENTAGE OF SALES BY PIZZA SIZE:

Generate a pie chart that represents the percentage of sales attributed to different pizza sizes.This chart will help us to understand customer preference for pizza sizes and their impact on sales.

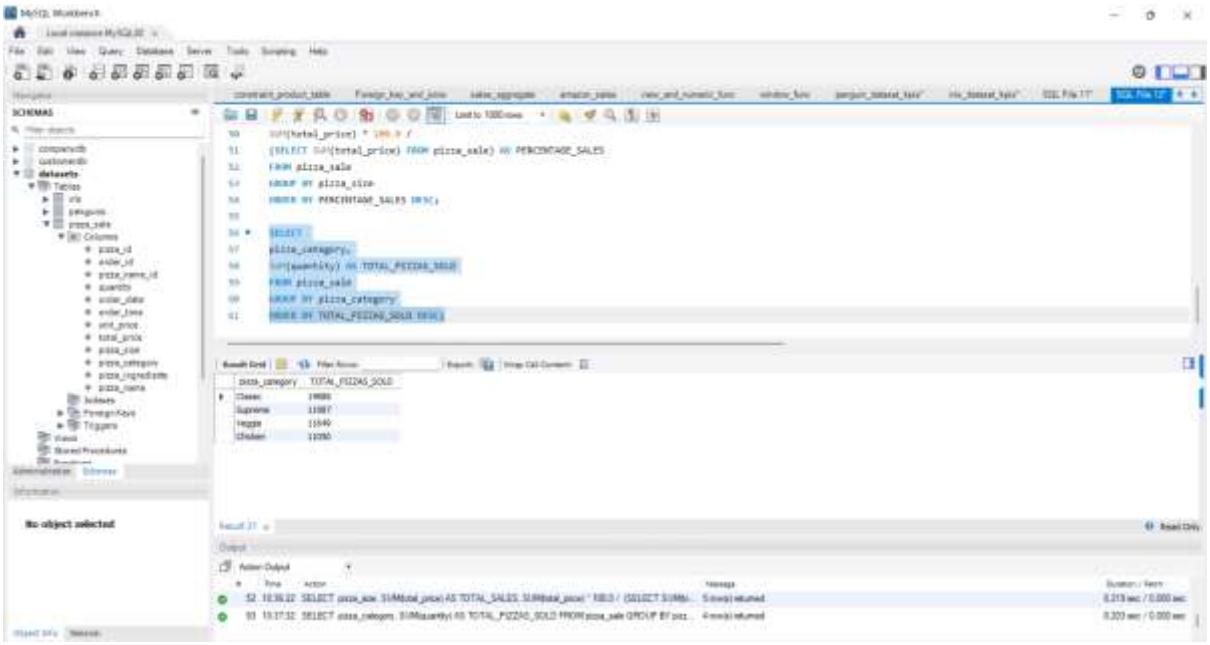
```
SELECT
pizza_size,
SUM(total_price) AS TOTAL_SALES,
SUM(total_price) * 100.0 /
(SELECT SUM(total_price) FROM pizza_sale) AS PERCENTAGE_SALES
FROM pizza_sale
GROUP BY pizza_size
ORDER BY PERCENTAGE_SALES DESC;
```



5.TOTAL PIZZA SOLD BY PIZZA CATEGORY:

Create a funnel chart that presents the total number of pizzas sold for each pizza category. This Chart will allow you to compare the sales performance of different pizza categories.

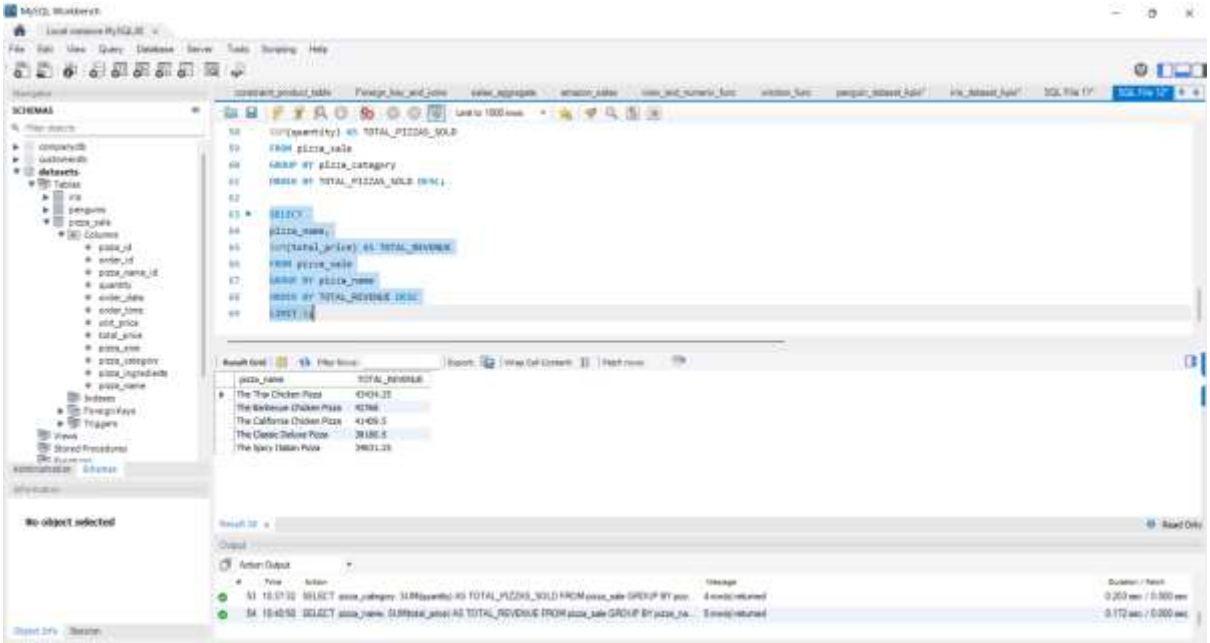
```
SELECT
pizza_category,
SUM(quantity) AS TOTAL_PIZZAS_SOLD
FROM pizza_sale
GROUP BY pizza_category
ORDER BY TOTAL_PIZZAS_SOLD DESC;
```



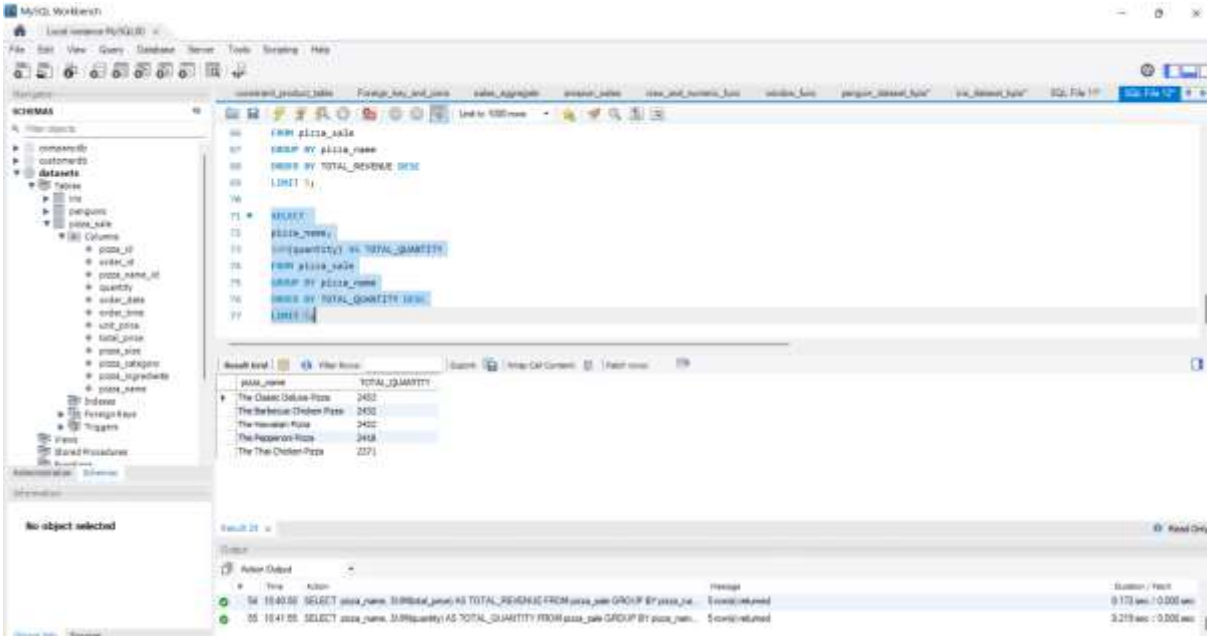
6.TOP 5 BEST SELLERS BY REVENUE, TOTAL QUANTITY AND TOTAL ORDERS:

Create A Bar chart highlighting the top 5 bestselling pizzas based on the revenue, total quantity, total orders. This chart will allow you to identify the most popular pizza options.

```
SELECT
pizza_name,
SUM(total_price) AS TOTAL_REVENUE
FROM pizza_sale
GROUP BY pizza_name
ORDER BY TOTAL_REVENUE DESC
LIMIT 5;
```

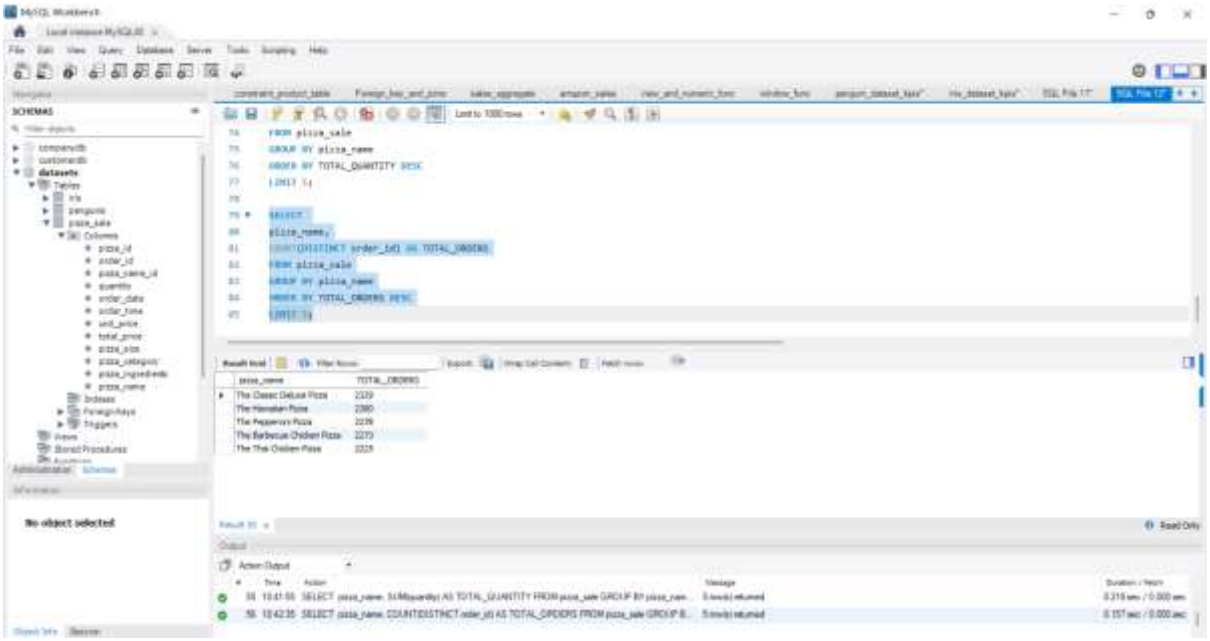


SELECT
pizza_name,
SUM(quantity) AS TOTAL_QUANTITY
FROM pizza_sale
GROUP BY pizza_name
ORDER BY TOTAL_QUANTITY DESC
LIMIT 5;



SELECT
pizza_name,
COUNT(DISTINCT order_id) AS TOTAL_ORDERS
FROM pizza_sale
GROUP BY pizza_name

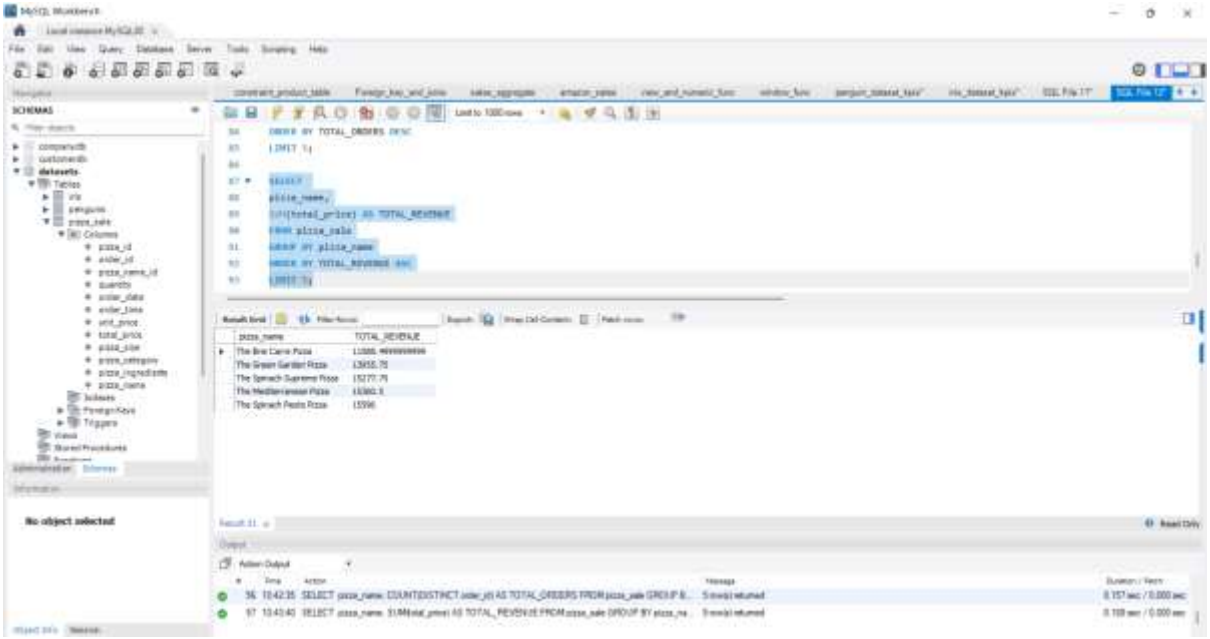
ORDER BY TOTAL_ORDERS DESC
LIMIT 5;



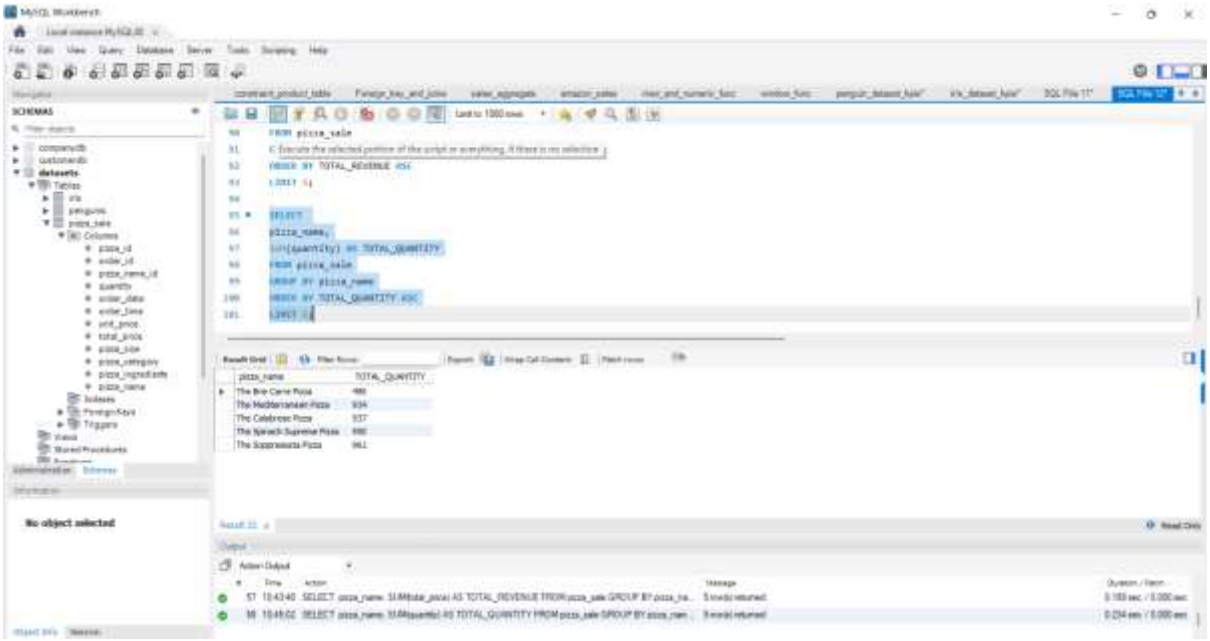
7.BOTTOM 5 BEST SELLERS BY REVENUE, TOTAL QUANTITY AND TOTAL ORDERS:

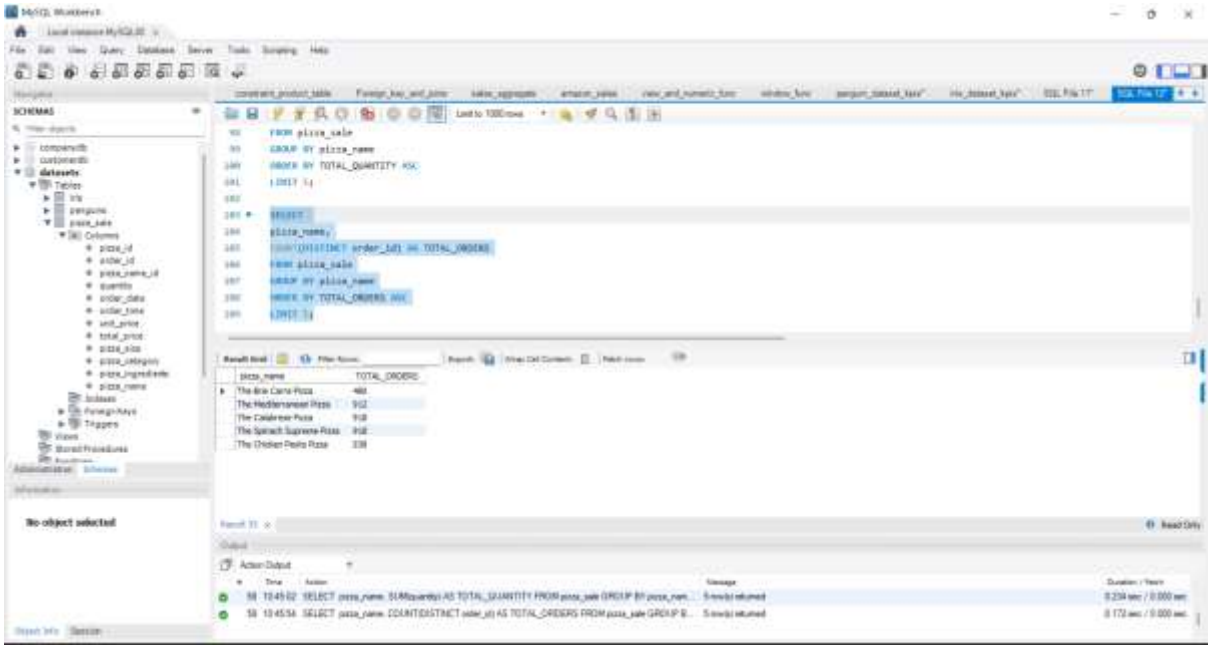
Create a bar chart showcasing the bottom 5 worst selling pizzas based on the revenue total quantity, total orders. This chart will enable us to identify underperforming or less popular pizza options.

```
SELECT
pizza_name,
SUM(total_price) AS TOTAL_REVENUE
FROM pizza_sale
GROUP BY pizza_name
ORDER BY TOTAL_REVENUE ASC
LIMIT 5;
```



SELECT
pizza_name,
SUM(quantity) AS TOTAL_QUANTITY
FROM pizza_sale
GROUP BY pizza_name
ORDER BY TOTAL_QUANTITY ASC
LIMIT 5;



SELECT
pizza_name,
COUNT(DISTINCT order_id) AS TOTAL_ORDERS
FROM pizza_sale
GROUP BY pizza_name

ORDER BY TOTAL_ORDERS ASC
LIMIT 5;